Experiment 1: Class Fundamentals

Program 1.1: Student Information System

```
class Student {
    int rollNo;
    String name;
    String branch;
    double cgpa;
    void display() {
        System.out.println("Roll Number: " + rollNo);
        System.out.println("Name: " + name);
        System.out.println("Branch: " + branch);
        System.out.println("CGPA: " + cgpa);
    }
}
public class StudentInfo {
    public static void main(String[] args) {
        Student s = new Student();
        s.rollNo = 101;
        s.name = "Raina";
        s.branch = "CSE-AIML";
        s.cgpa = 9.2;
        s.display();
    }
}
```

Output:

```
Roll Number: 101
Name: Raina
Branch: CSE-AIML
CGPA: 9.2
```

Program 1.2: Employee Information System

```
class Employee {
    int employeeId;
    String name;
    String department;
    double salary;
    void display() {
        System.out.println("Employee ID: " + employeeId);
        System.out.println("Name: " + name);
        System.out.println("Department: " + department);
        System.out.println("Salary: " + salary);
    }
}
public class EmployeeInfo {
    public static void main(String[] args) {
        Employee e = new Employee();
        e.employeeId = 101;
        e.name = "Raina";
        e.department = "IT";
        e.salary = 50000;
        e.display();
    }
}
```

Output:

```
Employee ID: 101
Name: Raina
Department: IT
Salary: 50000.0
```

Experiment 2: Declaring Objects

Program 2.1: Book Details

```
class Book {
    String title;
    String author;
    double price;
    void display() {
        System.out.println("Title: " + title);
        System.out.println("Author: " + author);
        System.out.println("Price: " + price);
        System.out.println();
    }
}
public class BookDetails {
    public static void main(String[] args) {
        Book b1 = new Book();
        Book b2 = new Book();
        Book b3 = new Book();
        b1.title = "Java Programming";
        b1.author = "James Gosling";
        b1.price = 500;
        b2.title = "Python Basics";
        b2.author = "John Smith";
        b2.price = 400;
        b3.title = "Data Structures";
        b3.author = "Mark Allen";
        b3.price = 600;
        b1.display();
        b2.display();
        b3.display();
```

```
    }
}
```

Output:

```
Title: Java Programming
Author: James Gosling
Price: 500.0
Title: Python Basics
Author: John Smith
Price: 400.0
Title: Data Structures
Author: Mark Allen
Price: 600.0
```

Program 2.2: Array of Student Objects

```
import java.util.Scanner;
class Student {
    int rollNo;
    String name;
    String branch;
    void display() {
        System.out.println("Roll Number: " + rollNo);
        System.out.println("Name: " + name);
        System.out.println("Branch: " + branch);
        System.out.println();
    }
}
public class StudentArray {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        Student[] students = new Student[5];
```

```
        for (int i = 0; i < 5; i++) {
            students[i] = new Student();
            System.out.println("Enter details of Student " + (i + 1));
            System.out.print("Roll Number: ");
            students[i].rollNo = sc.nextInt();
            sc.nextLine();
            System.out.print("Name: ");
            students[i].name = sc.nextLine();
            System.out.print("Branch: ");
            students[i].branch = sc.nextLine();
        }
        System.out.println("\nStudent Details:");
        for (int i = 0; i < 5; i++) {
            students[i].display();
        }
        sc.close();
    }
}
```

Output:

```
Enter details of Student 1
Roll Number: 101
Name: Raina
Branch: CSE-AIML
Enter details of Student 2
Roll Number: 102
Name: Rinku
Branch: CSE
Enter details of Student 3
Roll Number: 103
Name: Ananya
```

Branch: CSE-AIML

Enter details of Student 4

Roll Number: 104

Name: Kiran

Branch: ECE

Enter details of Student 5

Roll Number: 105

Name: Priya

Branch: IT

Student Details:

Roll Number: 101

Name: Raina

Branch: CSE-AIML

Roll Number: 102

Name: Rinku

Branch: CSE

Roll Number: 103

Name: Ananya

Branch: CSE-AIML

Roll Number: 104

Name: Kiran

Branch: ECE

Roll Number: 105

Name: Priya

Branch: IT

Experiment 3: Assigning Object Reference Variables

Program 3.1: Reference Assignment

```
class Student {
    String name;
```

```
    void display() {
        System.out.println("Name: " + name);
    }
}
public class ReferenceAssignment {
    public static void main(String[] args) {
        Student s1 = new Student();
        s1.name = "Raina";
        Student s2 = s1;
        System.out.println("Before modification:");
        s2.display();
        s1.name = "Rinku";
        System.out.println("After modification:");
        s2.display();
    }
}
```

Output:

Before modification:

Name: Raina

After modification:

Name: Rinku

Program 3.2: Comparing Object References

```
class Employee {
    int id;
    String name;
    Employee(int id, String name) {
        this.id = id;
        this.name = name;
    }
```

```
}
public class CompareReferences {
   public static void main(String[] args) {
      Employee e1 = new Employee(101, "Rahul");
      Employee e2 = e1;
      if (e1 == e2) {
         System.out.println("Both references point to the same object.");
      } else {
         System.out.println("References point to different objects.");
      }
   }
}
```

Output:

Both references point to the same object.

Experiment 4: Introducing Methods

Program 4.1: Calculator Using Methods

```
class Calculator {
   int add(int a, int b) {
      return a + b;
   }
   int subtract(int a, int b) {
      return a - b;
   }
   int multiply(int a, int b) {
      return a * b;
   }
   double divide(int a, int b) {
      return (double) a / b;
   }
```

```
}
public class CalculatorDemo {
    public static void main(String[] args) {
        Calculator c = new Calculator();
        int a = 20;
        int b = 10;
        System.out.println("Addition: " + c.add(a, b));
        System.out.println("Subtraction: " + c.subtract(a, b));
        System.out.println("Multiplication: " + c.multiply(a, b));
        System.out.println("Division: " + c.divide(a, b));
    }
}
```

Output:

Addition: 30

Subtraction: 10

Multiplication: 200

Division: 2.0

Program 4.2: Rectangle Operations

```
class Rectangle {
    double length;
    double width;
    double area() {
        return length * width;
    }
    double perimeter() {
        return 2 * (length + width);
    }
    void display() {
        System.out.println("Area: " + area());
        System.out.println("Perimeter: " + perimeter());
```

```
    }
}
public class RectangleDemo {
    public static void main(String[] args) {
        Rectangle r = new Rectangle();
        r.length = 10;
        r.width = 5;
        r.display();
    }
}
```

Output:

```
Area: 50.0
Perimeter: 30.0
```

Experiment 5: Constructors

Program 5.1: Student Constructors

```
class Student {
    String name;
    int rollNo;
    Student() {
        name = "Unknown";
        rollNo = 0;
    }
    Student(String name, int rollNo) {
        this.name = name;
        this.rollNo = rollNo;
    }
    void display() {
        System.out.println("Name: " + name);
        System.out.println("Roll Number: " + rollNo);
    }
```

```
}
public class StudentConstructor {
    public static void main(String[] args) {
        Student s1 = new Student();
        Student s2 = new Student("Raina", 101);
        System.out.println("Default Constructor:");
        s1.display();
        System.out.println("\nParameterized Constructor:");
        s2.display();
    }
}
```

Output:

```
Default Constructor:
Name: Unknown
Roll Number: 0
Parameterized Constructor:
Name: Raina
Roll Number: 101
```

Program 5.2: Constructor Overloading

```
class BankAccount {
    int accountNo;
    String name;
    double balance;
    BankAccount() {
        accountNo = 0;
        name = "Unknown";
        balance = 0;
    }
    BankAccount(int accountNo, String name) {
        this.accountNo = accountNo;
```

```
        this.name = name;
        balance = 0;
    }
    BankAccount(int accountNo, String name, double balance) {
        this.accountNo = accountNo;
        this.name = name;
        this.balance = balance;
    }
    void display() {
        System.out.println("Account Number: " + accountNo);
        System.out.println("Name: " + name);
        System.out.println("Balance: " + balance);
        System.out.println();
    }
}
public class BankAccountDemo {
    public static void main(String[] args) {
        BankAccount b1 = new BankAccount();
        BankAccount b2 = new BankAccount(101, "Raina");
        BankAccount b3 = new BankAccount(102, "Rinku", 25000);
        b1.display();
        b2.display();
        b3.display();
    }
}
```

Output:

```
Account Number: 0
Name: Unknown
Balance: 0.0
Account Number: 101
```

Name: Raina

Balance: 0.0

Account Number: 102

Name: Rinku

Balance: 25000.0

Experiment 6: The this Keyword

Program 6.1: Using this Keyword

```
class Student {
    String name;
    int age;
    Student(String name, int age) {
        this.name = name;
        this.age = age;
    }
    void display() {
        System.out.println("Name: " + this.name);
        System.out.println("Age: " + this.age);
    }
}
public class ThisDemo {
    public static void main(String[] args) {
        Student s = new Student("Raina", 19);
        s.display();
    }
}
```

Output:

Name: Raina

Age: 19

Program 6.2: Constructor Chaining

```
class Student {
    String name;
    int age;
    String branch;
    Student() {
        this("Unknown");
    }
    Student(String name) {
        this(name, 0);
    }
    Student(String name, int age) {
        this(name, age, "CSE");
    }
    Student(String name, int age, String branch) {
        this.name = name;
        this.age = age;
        this.branch = branch;
    }
    void display() {
        System.out.println("Name: " + name);
        System.out.println("Age: " + age);
        System.out.println("Branch: " + branch);
    }
}
public class ConstructorChaining {
    public static void main(String[] args) {
        Student s = new Student("Raina", 19, "CSE-AIML");
        s.display();
    }
```

```
}
```

Output:

Name: Raina

Age: 19

Branch: CSE-AIML

Experiment 7: Garbage Collection

Program 7.1: Demonstrating Garbage Collection

```
class Demo {
    int id;
    Demo(int id) {
        this.id = id;
    }
    protected void finalize() {
        System.out.println("Object " + id + " is garbage collected.");
    }
}
public class GarbageCollectionDemo {
    public static void main(String[] args) {
        Demo d1 = new Demo(1);
        Demo d2 = new Demo(2);
        Demo d3 = new Demo(3);
        d1 = null;
        d2 = null;
        d3 = null;
        System.gc();
        System.out.println("Garbage collection requested.");
    }
}
```

Output:

Garbage collection requested.

Program 7.2: Object Eligibility for Garbage Collection

```
class Demo {
    int id;
    Demo(int id) {
        this.id = id;
    }
}
public class ObjectEligibility {
    public static void main(String[] args) {
        Demo d1 = new Demo(1);
        d1 = null;
        Demo d2 = new Demo(2);
        d2 = new Demo(3);
        new Demo(4);
        System.gc();
        System.out.println("Objects are eligible for garbage collection.");
    }
}
```

Output:

Objects are eligible for garbage collection.

Experiment 8: Method Overloading

Program 8.1: Overloading Arithmetic Methods

```
class Calculator {
    int add(int a, int b) {
        return a + b;
    }
    int add(int a, int b, int c) {
        return a + b + c;
    }
```

Program 7.2: Object Eligibility for Garbage Collection

```
class Demo {
    int id;
    Demo(int id) {
        this.id = id;
    }
}
public class ObjectEligibility {
    public static void main(String[] args) {
        Demo d1 = new Demo(1);
        d1 = null;
        Demo d2 = new Demo(2);
        d2 = new Demo(3);
        new Demo(4);
        System.gc();
        System.out.println("Objects are eligible for garbage collection.");
    }
}
```

Output:

Objects are eligible for garbage collection.

Experiment 8: Method Overloading

Program 8.1: Overloading Arithmetic Methods

```
class Calculator {
    int add(int a, int b) {
        return a + b;
    }
    int add(int a, int b, int c) {
        return a + b + c;
    }
```

```
    double add(double a, double b) {
        return a + b;
    }
}
public class ArithmeticOverloading {
    public static void main(String[] args) {
        Calculator c = new Calculator();
        System.out.println("Two integers: " + c.add(10, 20));
        System.out.println("Three integers: " + c.add(10, 20, 30));
        System.out.println("Two doubles: " + c.add(10.5, 20.5));
    }
}
```

Output:

Two integers: 30

Three integers: 60

Two doubles: 31.0

Program 8.2: Overloading Area Methods

```
class Area {
    double area(double radius) {
        return Math.PI * radius * radius;
    }
    int area(int length, int width) {
        return length * width;
    }
    int area(int side) {
        return side * side;
    }
}
public class AreaOverloading {
```

```
    public static void main(String[] args) {
        Area a = new Area();
        System.out.println("Circle Area: " + a.area(5.0));
        System.out.println("Rectangle Area: " + a.area(10, 5));
        System.out.println("Square Area: " + a.area(4));
    }
}
```

Output:

Circle Area: 78.53981633974483

Rectangle Area: 50

Square Area: 16

Experiment 9: Using Objects as Parameters

Program 9.1: Passing Student Object

```
class Student {
    String name;
    int rollNo;
    Student(String name, int rollNo) {
        this.name = name;
        this.rollNo = rollNo;
    }
}
public class StudentParameter {
    static void display(Student s) {
        System.out.println("Name: " + s.name);
        System.out.println("Roll Number: " + s.rollNo);
    }
    public static void main(String[] args) {
        Student s = new Student("Raina", 101);
        display(s);
```

```
    }
}
```

Output:

Name: Raina

Roll Number: 101

Program 9.2: Comparing Employee Salaries

```
class Employee {
    String name;
    double salary;
    Employee(String name, double salary) {
        this.name = name;
        this.salary = salary;
    }
}
public class EmployeeSalary {
    static void compare(Employee e1, Employee e2) {
        if (e1.salary > e2.salary) {
            System.out.println(e1.name + " has the higher salary.");
        } else if (e2.salary > e1.salary) {
            System.out.println(e2.name + " has the higher salary.");
        } else {
            System.out.println("Both employees have the same salary.");
        }
    }
    public static void main(String[] args) {
        Employee e1 = new Employee("Rahul", 50000);
        Employee e2 = new Employee("Ananya", 60000);
        compare(e1, e2);
    }
```

```
}
```

Output:

Ananya has the higher salary.

Experiment 10: Returning Objects

Program 10.1: Returning a Student Object

```
class Student {
    String name;
    int rollNo;
    Student(String name, int rollNo) {
        this.name = name;
        this.rollNo = rollNo;
    }
    void display() {
        System.out.println("Name: " + name);
        System.out.println("Roll Number: " + rollNo);
    }
}
public class ReturnStudent {
    static Student createStudent() {
        Student s = new Student("Raina", 101);
        return s;
    }
    public static void main(String[] args) {
        Student s = createStudent();
        s.display();
    }
}
```

Output:

Name: Raina

Roll Number: 101

Program 10.2: Returning a Bank Account Object

```
class BankAccount {
    String name;
    double balance;
    BankAccount(String name, double balance) {
        this.name = name;
        this.balance = balance;
    }
    void display() {
        System.out.println("Name: " + name);
        System.out.println("Balance: " + balance);
    }
}
public class ReturnBankAccount {
    static BankAccount updateBalance(BankAccount account) {
        account.balance += 5000;
        return account;
    }
    public static void main(String[] args) {
        BankAccount b = new BankAccount("Ravi", 20000);
        b = updateBalance(b);
        b.display();
    }
}
```

Output:

Name: Ravi

Balance: 25000.0

Experiment 11: Understanding static

Program 11.1: Static Variable

```
class Student {
    String name;
    static int count = 0;
    Student(String name) {
        this.name = name;
        count++;
    }
    void display() {
        System.out.println("Name: " + name);
    }
}
public class StaticVariable {
    public static void main(String[] args) {
        Student s1 = new Student("Raina");
        Student s2 = new Student("Rinku");
        Student s3 = new Student("Amar");
        s1.display();
        s2.display();
        s3.display();
        System.out.println("Total Students: " + Student.count);
    }
}
```

Output:

Name: Raina

Output:

Name: Ravi

Balance: 25000.0

Experiment 11: Understanding static

Program 11.1: Static Variable

```
class Student {
    String name;
    static int count = 0;
    Student(String name) {
        this.name = name;
        count++;
    }
    void display() {
        System.out.println("Name: " + name);
    }
}
public class StaticVariable {
    public static void main(String[] args) {
        Student s1 = new Student("Raina");
        Student s2 = new Student("Rinku");
        Student s3 = new Student("Amar");
        s1.display();
        s2.display();
        s3.display();
        System.out.println("Total Students: " + Student.count);
    }
}
```

Output:

Name: Raina

Name: Rinku

Name: Amar

Total Students: 3

Program 11.2: Static Methods

```
class Utility {
    static int square(int n) {
        return n * n;
    }
    static int cube(int n) {
        return n * n * n;
    }
    static long factorial(int n) {
        long fact = 1;
        for (int i = 1; i <= n; i++) {
            fact *= i;
        }
        return fact;
    }
}
public class StaticMethods {
    public static void main(String[] args) {
        int n = 5;
        System.out.println("Square: " + Utility.square(n));
        System.out.println("Cube: " + Utility.cube(n));
        System.out.println("Factorial: " + Utility.factorial(n));
    }
}
```

Output:

Square: 25

Cube: 125

Factorial: 120

Experiment 12: Introducing final

Program 12.1: Final Keyword

```
class Parent {
    final int VALUE = 100;
    final void display() {
        System.out.println("This is a final method.");
        System.out.println("Final variable: " + VALUE);
    }
}
final class FinalClass {
    void show() {
        System.out.println("This is a final class.");
    }
}
public class FinalDemo {
    public static void main(String[] args) {
        Parent p = new Parent();
        p.display();
        FinalClass f = new FinalClass();
        f.show();
    }
}
```

Output:

This is a final method.

Final variable: 100

This is a final class.

Program 12.2: Blank Final Variable

```
class Employee {
    final int employeeId;
    String name;
    Employee(int employeeId, String name) {
        this.employeeId = employeeId;
        this.name = name;
    }
    void display() {
        System.out.println("Employee ID: " + employeeId);
        System.out.println("Name: " + name);
    }
}
public class BlankFinal {
    public static void main(String[] args) {
        Employee e1 = new Employee(101, "Raina");
        Employee e2 = new Employee(102, "Rinku");
        e1.display();
        System.out.println();
        e2.display();
    }
}
```

Output:

Employee ID: 101

Name: Raina

Employee ID: 102

Name: Rinku

Experiment 13:

Program 13.1: College and Department

```
class College {
    static class Department {
        String name;
        Department(String name) {
            this.name = name;
        }
        void display() {
            System.out.println("Department: " + name);
        }
    }
}
public class CollegeDemo {
    public static void main(String[] args) {
        College.Department d =
                new College.Department("CSE-AIML");
        d.display();
    }
}
```

Output:

Department: CSE-AIML

Program 13.2: Employee Address

```
class Employee {
    int id;
```

```
    String name;
    Employee(int id, String name) {
        this.id = id;
        this.name = name;
    }
    static class Address {
        String city;
        String state;
        Address(String city, String state) {
            this.city = city;
            this.state = state;
        }
        void display() {
            System.out.println("City: " + city);
            System.out.println("State: " + state);
        }
    }
}
public class EmployeeAddress {
    public static void main(String[] args) {
        Employee e = new Employee(101, "Ravi");
        Employee.Address address =
                new Employee.Address("Vijayawada", "Andhra Pradesh");
        System.out.println("Employee ID: " + e.id);
        System.out.println("Name: " + e.name);
        address.display();
    }
}
```

Output:

Employee ID: 101

Name: Ravi

City: Vijayawada

State: Andhra Pradesh

Experiment 14: Inner Classes

Program 14.1: Student Address

```
class Student {
    String name;
    int rollNo;
    Student(String name, int rollNo) {
        this.name = name;
        this.rollNo = rollNo;
    }
    class Address {
        String city;
        String state;
        Address(String city, String state) {
            this.city = city;
            this.state = state;
        }
        void display() {
            System.out.println("Student Name: " + name);
            System.out.println("Roll Number: " + rollNo);
            System.out.println("City: " + city);
            System.out.println("State: " + state);
        }
    }
}
```

```
public class StudentAddress {
    public static void main(String[] args) {
        Student s = new Student("Ravi", 101);
        Student.Address a =
                s.new Address("Vijayawada", "Andhra Pradesh");
        a.display();
    }
}
```

Output:

Student Name: Ravi

Roll Number: 101

City: Vijayawada

State: Andhra Pradesh

Program 14.2: Library Management

```
class Library {
    String libraryName;
    Library(String libraryName) {
        this.libraryName = libraryName;
    }
    class Book {
        String title;
        String author;
        Book(String title, String author) {
            this.title = title;
            this.author = author;
        }
        void display() {
            System.out.println("Library: " + libraryName);
            System.out.println("Book Title: " + title);
```

```
            System.out.println("Author: " + author);
        }
    }
}
public class LibraryManagement {
    public static void main(String[] args) {
        Library library = new Library("Central Library");
        Library.Book book =
                library.new Book("Java Programming", "James Gosling");
        book.display();
    }
}
```

Output:

Library: Central Library

Book Title: Java Programming

Author: James Gosling